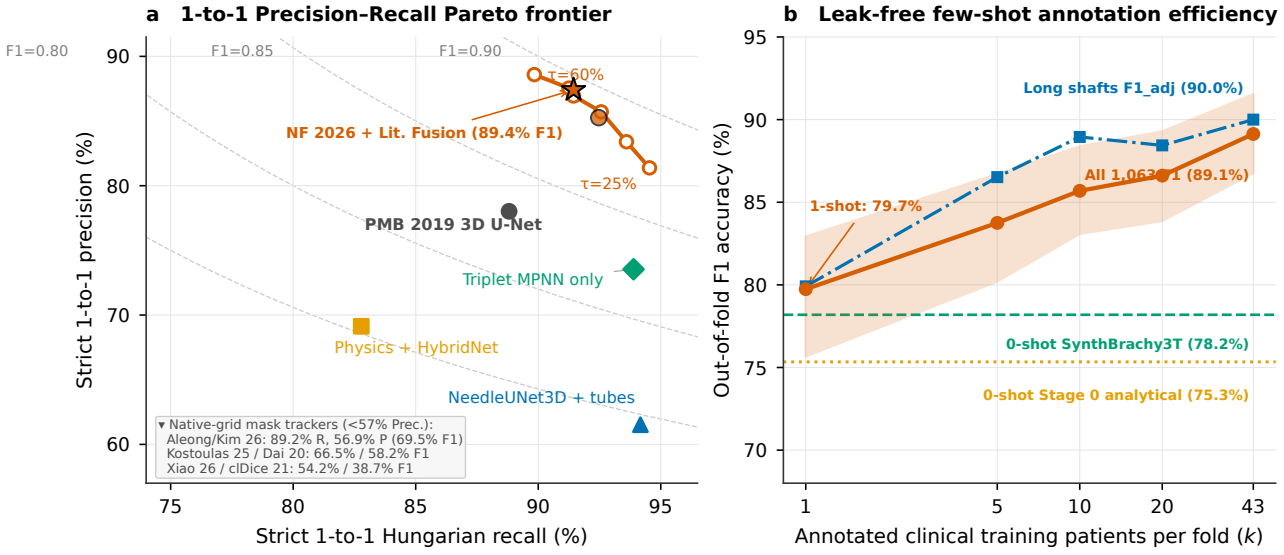
a 1-to-1 Precision–Recall Pareto frontier
F1=0.80
F1=0.85
F1=0.90
τ=60%
NF 2026 + Lit. Fusion (89.4% F1)
τ=25%
PMB 2019 3D U-Net
Triplet MPNN only
Physics + HybridNet
NeedleUNet3D + tubes
▾ Native-grid mask trackers (<57% Prec.):
Aleong/Kim 26: 89.2% R, 56.9% P (69.5% F1)
Kostoulas 25 / Dai 20: 66.5% / 58.2% F1
Xiao 26 / clDice 21: 54.2% / 38.7% F1
Strict 1-to-1 precision (%)
Strict 1-to-1 Hungarian recall (%)
b Leak-free few-shot annotation efficiency
Long shafts F1_adj (90.0%)
All 1,063 F1 (89.1%)
1-shot: 79.7%
0-shot SynthBrachy3T (78.2%)
0-shot Stage 0 analytical (75.3%)
Out-of-fold F1 accuracy (%)
Annotated clinical training patients per fold (k)